

A Scoping review of PubMed studies prior to 2026 on the applications of kinetic models in systems biology

Abstract

Kinetic model is a mathematical or computational model that has been developed and applied in many industries. Numerous kinetic models have been constructed and used to study different area of systems biology. The areas include modelling metabolic, enzymatic, signalling pathway, gene regulation, drug development and cellular behaviour. Hence, a scoping review was performed using PubMed for studies published prior to 2026. 128 articles out of 193 articles were included and classified into 2 main themes: (a) development of kinetic model, and (b) application of kinetic model. The majority of these papers focused on the development of kinetic model through constructing, adopting and modifying. Metabolism and enzymatic models are the most developed ones. Through this review, we could discover which type of kinetic models are available and which type of kinetic model requires further research or improvement which is an essential tool in studying systems biology.

Keywords: kinetic model, systems biology, applications

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 Yu Huan Mun,^{1,2} Maurice HT Ling^{2,3,4}
¹School of Health & Life Sciences, Teesside University, UK

²Management Development Institute of Singapore, Singapore

³Newcastle Australia Institute of Higher Education, University of Newcastle, Australia

⁴HOHY PTE LTD, Singapore

Correspondence: Maurice HT Ling, Management Development Institute of Singapore, Singapore

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Introduction

Systems biology is a complete and in-depth study of natural and synthetic molecular diversity of living system.¹⁻³ It involves the identification of general principles and patterns in biology systems and using this knowledge to develop kinetic models of regulatory network to foresee and explain the behaviour behind it.⁴ The aim of systems biology is not solely for describing life but to also comprehend and engineer it for the improvement of humanity.

Kinetic model is a type of mathematical model that explains the rate of change of variables over time.⁵ Kinetic models are developed as a tool in many fields such as chemistry, biology, pharmacology, environment science and engineering. Using biological data, kinetic models can be created to investigate physiological or pathophysiological states and predict biological outcomes.⁶ Kinetic models can be developed for numerous utilities such as studying gene regulatory network, signal transduction, signalling pathway,

metabolic pathway, protein interaction, tissue or whole cell organism interactions. A kinetic model could be simplified, extended or modified to fit the aim of the study. This scoping review examines the type of kinetic models developed, adopted or extended and how existing kinetic models are used in the study of systems biology.

Methods

A PubMed search was carried out on 01 February 2026 for existing studies regarding the application of kinetic models in systems biology published before 2026 using PRISMA guidelines;⁷ specifically, pertaining to scoping review.⁸ The search term “kinetic model” AND “systems biology” were used. This search term is intentionally restrictive which is appropriate, as shown in scoping reviews.⁹⁻¹¹ Date range limit was set between 01 January 1000 (the default start date in PubMed search when a start date is not given) and 01 January 2026. This resulted in the following reproducible search URL,¹²

[https://pubmed.ncbi.nlm.nih.model"+AND+"systems+biology"&filter=dates.1000/1/1-2026/1/1](https://pubmed.ncbi.nlm.nih.model).

The following exclusion criteria were used: (A) articles with no full text; (B) articles not in English; (C) non-primary articles; (D) articles not about creating or using kinetic model. Screening and theming of articles are initially done by the first author (YHM), followed by verification by the second author (ML).

Results

A total of 193 articles were identified from PubMed (Figure 1). After the screening process, 128 articles were included and segregated into 2 themes – (a) development of kinetic models, and (b) application

of kinetic models (Table 1). Nearly half of the articles identified were focused on developing kinetic models on metabolic and enzymatic pathway. In contrast, there were much lesser articles published on the development of kinetic model related to protein interaction. One-tenth of the included articles describe the use of published kinetic model for their respective studies. Due to the large number of included studies, representative references were cited with the remaining references listed as Supplementary materials. Details from the screening of articles; such as, which article has been excluded under which criteria; are also provided in Supplementary materials for the purpose of transparency as previously proposed.¹²

Table 1 Thematic classification of articles on development and application of kinetic models

Theme	Types of Kinetic Model	Number of Articles with Representative Articles	Percentage
Development of kinetic model (including adopting and modifying of kinetic model)	Metabolic / Enzymatic Pathway	58 ^{13–16}	45.31%
	Gene Regulation	12 ^{17–20}	9.38%
	Signalling	22 ^{21–24}	17.19%
	Protein Interaction	7 ^{25–28}	5.47%
	Integrated Cellular	8 ^{29–32}	6.25%
Using of published kinetic models	Others	8 ^{33–35}	6.25%
	-	13 ^{36–40}	10.16%

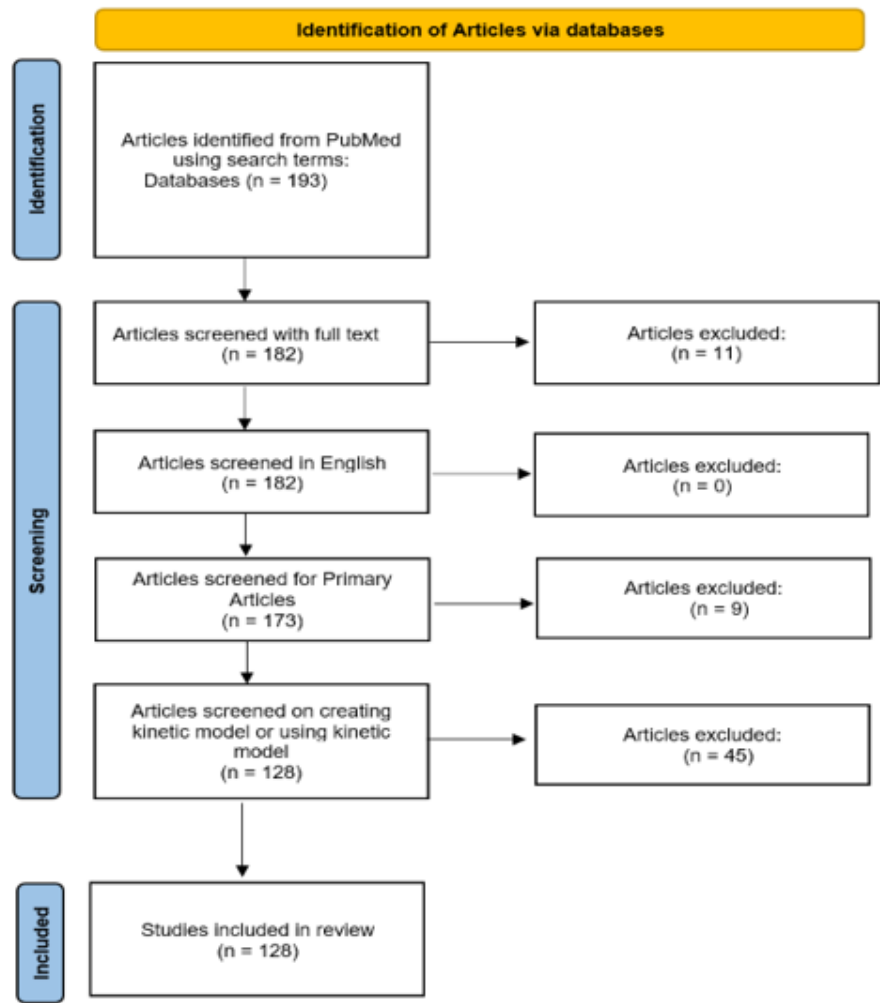


Figure 1 Articles selection process using PRISMA for Scoping Reviews.^{7,8}

There are no overlaps in themes as the total number of studies across all themes and subthemes add up to the 128 included studies.

Theme 1: Development of kinetic model

There were various types of kinetic model that were developed for many purposes. The screened articles are further classified based on the various types of kinetic model created in the study. The 5 main groups of kinetic models include metabolic enzymatic, gene regulatory, signalling, protein interaction and integrated cellular. The

rest of the kinetic models that did not fit into the 5 main groups were classified as “Others”.

Theme 1.1: Development of kinetic model – metabolic and enzymatic kinetic models

There was a total of 58 articles included for the development of kinetic models on metabolic and enzymatic pathway. Most of the models were deterministic kinetic model to study metabolic and enzymatic pathway. The different organisms that were investigated

include plants, human cells, bacteria and yeast. Central carbon metabolism and glycolysis process had a higher proportion of being modelled and studied. Parasite, bacterial and yeast metabolisms were heavily studied. Common organism used include *Plasmodium falciparum*, *Escherichia coli*, pancreatic beta cells, and *Saccharomyces cerevisiae*.

A study developed kinetic model on the intracellular central carbon metabolism in pancreatic beta cells when stimulated with high levels of glucose.⁴¹ The model includes 56 metabolites and 65 enzymes catalysed reactions. The model studied the relationship between beta cell metabolism and glucose stimulated insulin secretion. It helps to predict the effectiveness of diabetic medication on insulin secretion. A study constructed a detailed kinetic model on the central carbon metabolism of *E. coli* in a batch culture. The model includes 27 metabolites, 38 fluxes, 21 gene expressions, 12 biomass rate equation, 22 enzymes and phosphotransferase system proteins.¹⁴

A study constructed a detailed mechanistic kinetic model with 19 metabolites and 17 reactions on trophozoite stage of *P. falciparum* glycolysis.¹⁵ The model was validated and was used to predict parasite metabolism. Another study modelled glucose metabolism in malaria patients to investigate the contribution of parasite metabolism to the disease state.⁴² The kinetic model includes 35 to 45 reactions and 30 to 40 metabolites. The primary organism used was *P. falciparum* and human red blood cell was used as the host cell.

Diabetes, red blood cells metabolism, inflammation disease, infections and cancers linked to metabolic modification were studied with the development of kinetic models. By studying the affected pathways, it helps to study the mechanism behind the diseases and ways to develop novel drugs. Kinetic model used to study the disease mechanism helps to better understand pathological and pharmacological states. A study developed a kinetic model of arachidonic acid metabolism pathway in non-alcoholic fatty liver disease (NAFLD).⁴³ A/J, C57BL/6J, and PWD/PhJ were the 3 mouse strains used in the study for the comparison of the severity of steatohepatitis.

Lipid metabolisms such as phospholipid synthetic pathway, fatty acid beta oxidation and sphingolipid biosynthesis were analysed to gain insights on their physiological roles. A study builds a kinetic model on glycerophospholipid pathway in *Plasmodium knowlesi*.⁴⁴ The model studied how the pathway metabolism and the resistance mechanism were affected during a malaria infection.

To study how carbohydrates were broken down and regulated in the cell, kinetic models were constructed for sucrose, fructose, lactose metabolism. Several studies also focused on redox metabolism and homeostasis in cells such as hydrogen peroxide metabolism, copper homeostasis and glutathione metabolism. Redox metabolism is important in explaining how cells utilise energy and assist in the diagnosis and treatment of diseases. A study developed a kinetic model on copper homeostasis in *S. cerevisiae*. The model includes 25 reactions and 10 components in the cytosol of *S. cerevisiae* used to study yeast mechanism when the copper concentration increases.¹³

Apart from human biology, plant metabolic kinetic models were developed to help in the study of plant metabolism, biosynthesis and photosynthesis. To study the physical and chemical process of sucrose cycling, a kinetic model was developed. The model investigated the subcellular sugar concentrations in *Arabidopsis thaliana* in various light and heat conditions.¹⁶ Six ODE equations were used to describe the time dependent change.

Theme 1.2: Development of kinetic model – gene regulatory kinetic models

12 articles developed kinetic models on gene regulatory. Most developed a deterministic ordinary differential equation while the minority developed a stochastic model. A kinetic model was created in plant biology in the study of regulation of ethylene synthesis during the development and ripening of tomato.⁴⁵ Gene expression was used in the prediction of intermediate protein and metabolites in ethylene biosynthesis pathway.

Three articles developed kinetic models relating to cell differentiation which include kinetic model of multi-miRNA chondrogenesis, B-cells differentiation to plasma and memory cells, and the interaction of transcription factor PU.1-GATA-1. The transcription factors PU.1 and GATA-1 regulate hematopoietic system.¹⁷ A gene regulatory network model that activates time dependent changes in gene expression and phenotypes affected by miRNAs and their targets was constructed.⁴⁶

Kinetic model was developed to predict chemoresistance in human cancer cells. Main molecules E2F1, p73, DNp73, miR-205, ERBB3, BCL2 were included in the development of the kinetic gene regulatory network model.²⁰ Some studies target the gene regulatory network on the feedback regulation and feed forward regulation. Target genes CTA1 and POT1 were included in the model of feed forward regulation of the core fatty acid transcriptional network in *S. cerevisiae*.¹⁹ Kinetic model of the Clb/Cdk1–Fkh2 regulatory system and Boolean regulatory model in *S. cerevisiae* was developed to study how CLB genes affect the expression in cell cycle and division. CLB2 and CLB3 were also included in the model.⁴⁷

To investigate the gene p21, an ODE gene regulatory model was created to predict p21 protein level for 9 cellular functions.¹⁸ 1 kinetic model studied the ATF3/NF- κ B regulation of IL6 and IL12b which explain ATF3 as a regulator of negative feedback that regulate immune activation after the stimulation of Toll-like receptor 4.⁴⁸ The included articles for gene regulatory model show that kinetic model was an effective way to study gene regulation and predicting behaviours of regulatory network in systems biology.

Theme 1.3: Development of kinetic model – signalling kinetic models

In the included articles, 22 models focused on signalling pathway. There was a wide range of biological signalling pathway. TGF- β signal transduction, IGF network activity, rapamycin (mTOR1/2), and phosphoinositide-3 kinase (PI3K) and neuregulin induced HER2/3 through MAPK and PI3K were models developed to study the respective growth factors.²² Insulin Receptor Substrate 1 (IRS-1) and Insulin Receptor Substrate 2 (IRS-2) are important proteins that signal insulin and growth factor to the cell. Deterministic ODE kinetic model was created to stimulate insulin signalling of liver cells in humans.²⁴ The model consists of 40 to 80 types of reactions and parameters respectively.

Four papers also created kinetic models to study the regulation and interaction of G protein coupled systems. G protein coupled reaction (GPCR) happens through sequential receptor reactions.⁴⁹ GPCR activates G protein. This explains the different pharmacology effect when different drugs bind to the same receptor. Real time signalling timing in GPCR expressing living cells was measured in the study. GTPase Activating Protein (GAP) are regulator protein that deactivate G protein. One of the studies explained the strength and timing of G protein signalling was closely regulated by GPCR and GAP.⁵⁰ The

coordination between GPCR and GAP is important in controlling the deactivation of G protein at the correct setting.⁵¹

Bacterial signalling studied include Two-Component System (TCS) signalling in *E. coli* and signalling processes in phototaxis and chemotaxis. A mechanistic kinetic model was created with 10 biochemical parameters to study the signalling pathway in *E. coli*.²³ The study predicts the behaviour of multiple gene knockout that removes buffering mechanism against cross talk. Both EnvZ/OmpR and CpxA/CpxR systems were included in the study. Another study constructed a stochastic kinetic model based on published data. The study shows that based on actual kinetic parameters, TCS demonstrate all-or-none, graded or mixed mode responses.⁵²

Other signalling pathway model includes platelet plasma ODE model on thrombin substrates Z-GGR-MCA and Boc-VPR-MCA. The model was built upon Hockin Mann model with 76 species, 56 reactions and 105 kinetic parameters to evaluate thrombin substrates that were commonly used.²¹ Signalling models for hormone estrogen receptor-a regulation in positive breast cancer cell, calcium signalling pathway, cellular JAK-STAT and PKD-CERT signalling and interaction were also developed.

Overall, signalling models had a higher ratio in developing models on growth factor, bacterial and G-protein coupled reaction protein signalling pathway.

Theme 1.4: Development of kinetic model – protein interaction kinetic models

There were seven included articles on the development of kinetic model on protein interaction. These models include multivalent sequential binding between recombinant Protein A/G (PAG) and Immunoglobulin G (IgG) which uses *Staphylococcus aureus* as the source for protein A and *Streptococcus* as source of protein G.²⁷ Ion permeation channel through the Kir4.1 potassium channel with K^+ , Rb^+ , Cs^+ , NH_4^+ , Ba^{2+} ions were studied.⁵³ Kir4.1 expressed in *Xenopus laevis* oocytes was tested in the kinetic model of the ion channel regulation.

Two kinetic models were developed for the study of DNA replication and Apoptosis-associated Speck-like (ASC) protein oligomerization.^{28,54} The kinetic model demonstrate ASC self-assembles faster and into more compact filaments than ASCb due to differences in linker length. Kinetic model of Protease-Activated Receptor (PAR-3) was developed to determine the membrane binding, oligomerization and dissociation in *Caenorhabditis elegans* zygote with PAR-3, PAR-6, PKC-3, CDC-42 to form the complex.²⁵ One of the kinetic models adopted Hill's dynamic instability model and improvised a new FRAP interpretation model.²⁶ It helped to interpret FRAP experimental data in terms of the underlying microtubule dynamic.

Stochastic kinetic model of a chromatophore vesicle from the purple bacterium *Rhodobacter sphaeroides* was extended from a published model.⁵⁵ The new model uses a stochastic protein-like kinetic framework to simulates photosynthetic energy conversion process in *Rhodobacter sphaeroides* chromatophore vesicles to investigate how molecular separation and stochasticity affected system behaviour.

Theme 1.5: Development of kinetic model – integrated cellular kinetic models

Eight articles regarding kinetic models were related to integrated cellular. Integrated cellular kinetic model combined

more than 1 biological process. One of the studies integrated three biological processes of Central metabolic pathways of hepatic glucose metabolism with insulin and glucagon signalling and gene regulation.³⁰ Other kinetic models integrated two biological processes such as building kinetic model with 15 to 30 reactions to simulate iNOS gene expression patterns and signal transduction in murine macrophages.³¹ One of the studies extended a kinetic model by replacing the intrinsic clearance term with a detailed kinetic model of estrogen metabolism in the liver or a genome-scale model of liver metabolism.³² Medical conditions studied related kinetic models include multi-organ and scale that express the reactions of enzymes and transporters with the regulation of hormonal actions in diabetes; sugar and fats metabolism in Non-Alcoholic Fatty Liver Disease; estrogen metabolism in liver and the interaction of immune system with melanoma micro metastasis. In plant biology, stimulation time of central cystolic carbohydrate metabolism was studied by developing a kinetic model of SnRK1/C-metabolism interaction during a repletion experiment of C-starved seedlings of *A. thaliana*.²⁹

Theme 1.6: Development of kinetic model – others

The remaining eight studies developed kinetic models to study about physiological, locomotion, viral replication of Eastern equine encephalitis life cycle, drugs toxicity in chemo immune system and GeXP multiplex RT-PCR assay calibration. In locomotion model, the movement, speed and motility of bacterial in biofilm and *C. elegans* were studied.^{33,35} In the viral replication model, it explains the whole process of how a virus infect a healthy cell.³⁴ The processes include attachment to receptor, entering the host cell, uncoating, RNA replication, assembly and lastly exiting the cell. These kinetic models show the importance of kinetic models in studying different biological and physiological processes.

Theme 2: Using of published kinetic model

Current kinetic models have a wide range of applications and could be used in numerous fields. It can be applied for identification of ligands targeting G protein coupled receptor and transcription factor target. In the study of G protein coupled receptor, the model helped to identify antagonists with novel chemical structures that would have been missed by traditional ligand-based approach and help to predicted dose-response and signalling inhibition.³⁷ Kinetic models are also used in studying the diagnosis, treatment of liver cancer and lifespan of red blood cell. Based on the study, standard laboratory HbA1c can be misleading as it does not account for differences in every individual red blood cell lifespan.⁴⁰ Kinetic models could help in investigating BRAF inhibitor drug tolerance, the relationship in intracellular metabolism, effect of RNA interference on the target of mRNA degradation, organism responses to light and temperature. One of the studies uses Fokker-Planck-type (FP-type) kinetic model to investigate how melanoma cancer cells develop resistance over time towards BRAF inhibitor drugs.³⁹ BRAF inhibitor drugs were used to treat melanoma and developing resistance could affect the effectiveness of the treatment. The study shows that the development of melanoma cell resistance was not only due to DNA mutation. It was instead due to the dynamic and unpredictable change in gene activity affecting the melanoma cell state. Hence, it is important to understand the transition state of the cell to understand the reason behind therapy failure. One of the studies used yeast glycolysis model to investigate how temperature affect the rate of glycolytic metabolism. The organism used in the study was *S. cerevisiae*. Based on the study, temperature was not the limiting factor on the rate of metabolism as the rate of enzyme catalytic change co-ordinately with temperature.³⁶ Glucose concentration was the main factor that affected

the rate of metabolism. Other applications also include validation in plant biology and helping in computing and verifying the rate of bacterial growth. Kinetic model of *E. coli* metabolism was used to study the relationship between glycolysis, acetate and rate of growth change.³⁸ The study shows that acetate was a useful nutrient that aids in bacterial growth. The presence of acetate helps *E. coli* to survive and grow despite a disruption in glycolysis. The included studies cover the wide range of uses of kinetic models from studying cellular to organism to therapeutic and validation applications.

Discussion

In this scoping review, 128 selected articles show that there were an extensive development, extension, modification and application of kinetic model in system biology. It was used to study, validate and experiment our biological system in terms of understanding our metabolic pathway, gene regulation, signalling processes and protein interactions. Applications of kinetic models were not limited to a few categories. It can also be used to study the movement of organism and understand the life cycle of virus replication. Applications of kinetic model is crucial in analyse the failure causes of cancer treatments and identify ways to reduce drug interactions and resistance. Numerous models were created to research about the glycolysis pathway in parasite which helps in anti-malarial drug targets identifications. Kinetic models also helped in better understanding of a disease or pathway which is essential for future therapeutic progression. With the use of kinetic models, more hypothesis and experiment could be tested out. Simulations and biological response to different factors could be predicted and studied.

About half of the developed kinetic models were on metabolic and enzymatic pathway. One of the possible explanations is that constructing a metabolic model could be easier to develop, measure, estimate and validate as compared to other types of kinetic model.⁵⁶ There were many established kinetic laws such as Michaelis-Menton kinetics, Mass-action kinetics and Hill kinetics which were used as the foundation of developing kinetic models. Reactions and enzymes were also well-researched. Some studies adopted published kinetic model and reconstruct a new updated kinetic model. The existing models could be extended or modified by including more reactions, metabolites or changing the parameter kinetics. One of the benefits of adopting published kinetic models is that these models were validated. Therefore, time was saved from having to build a kinetic model from scratch. On the other hand, gene regulatory and signalling models does not have a fixed kinetic framework. The pathways in gene regulation and signalling are more complex.⁵⁷ There were relatively lesser data and information available. Most models adapted parameters that were derived from other models. It was more difficult for kinetic parameters estimation, measurement and model validation. Hence, these models are more challenging to developed accurately as compared to metabolic models.

The strengths of this review are reviewing the type of kinetic models developed and its application using PRISMA guideline to aid in reproducibility, completeness, and transparency.⁷ The kinetic models are grouped based on their application and hence allow to review the trend of the constructed model. However, limitations of this review include reviewing articles only from PubMed. Articles were only screened based on the inclusion criteria and most of the models were not readily available for download. As kinetic models are progressing rapidly, articles published after the search period were not screened or included in this review.

The possible gaps in developing kinetic model might be due to the lack of accurate parameters and restricted model validation.⁵⁸ Future studies should focus on improving the accuracy of parameter estimation. Development of kinetic model could focus on least explored areas such as gene regulation, signalling pathway and protein interaction. Future studies could also apply kinetic models in emerging disease diagnoses, testing, treatments and experimenting biological responses.

Conclusion

Kinetic model is a mathematical model or computational model that demonstrate the rate of change in systems biology. It is constructed and applied in numerous areas to study biological reactions, gene regulation, signalling, protein interaction and physiological behaviour. This systematic review summarised the applications of kinetic model into 2 themes; development of kinetic models and application of existing kinetic model. Currently, there is an increasing trend in developing kinetic models for a study. Kinetic model remains an important tool in studying systems biology.

Supplementary materials

Supplementary materials can be downloaded from https://bit.ly/KM_SR.

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Conflict of interest

The authors declare no conflict of interest.

References

1. Kitano H. Systems biology: a brief overview. *Science*. 2002;295(5560):1662–1664.
2. Isalan M. Systems biology: A cell in a computer. *Nature*. 2012;488(7409):40–41.
3. Zheng X, Zheng P, Sun J. Systems biology for industrial biotechnology. *Chin J Biotechnol*. 2019;35(10):1955–1973.
4. Hussain S, Yates C, Campbell MJ. Vitamin D and Systems Biology. *Nutrients*. 2022;14(24):5197.
5. Kaltschmidt JA, Martinez Arias A. A new dawn for an old connection: development meets the cell. *Trends Cell Biol*. 2002;12(7):316–320.
6. Foster CJ, Wang L, Dinh HV, et al. Building kinetic models for metabolic engineering. *Curr Opin Biotechnol*. 2021;67:35–41.
7. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71.
8. Tricco AC, Lillie E, Zarin W, et al. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. *Ann Intern Med*. 2018;169(7):467–473.
9. Sítima G, Galhardo-Branco C, Reis-Pina P. Equity of access to palliative care: a scoping review. *Int J Equity Health*. 2024;23(1):248.
10. Silva I, Santos AC, Oliveira A. Disabling Norms, Affirming Desires: A Scoping Review on Disabled Women's Sexual Practices. *Societies*. 2025;15(6):154.

11. Wang W, Lu S, Tang H, et al. A scoping review of drug epidemic models. *Int J Environ Res Public Health*. 2022;19(4):2017.
12. Hon RY, Ling MH. Science/Education Portraits IX – Reproducibility and Transparency of Systematic Reviews. *Medicon Med Sci*. 2024;7(2):36–40.
13. Dulaney C, Walton JR, Lindahl PA. A kinetic model of copper homeostasis in *Saccharomyces cerevisiae*. *J Biol Chem*. 2025;301(8):110368.
14. Jahan N, Maeda K, Matsuoka Y, et al. Development of an accurate kinetic model for the central carbon metabolism of *Escherichia coli*. *Microb Cell Fact*. 2016;15(1):112.
15. Penkler G, du Toit F, Adams W, et al. Construction and validation of a detailed kinetic model of glycolysis in *Plasmodium falciparum*. *FEBS J*. 2015;282(8):1481–1511.
16. Weiszmann J, Fürtauer L, Weckwerth W, et al. Vacuolar sucrose cleavage prevents limitation of cytosolic carbohydrate metabolism and stabilizes photosynthesis under abiotic stress. *FEBS J*. 2018;285(21):4082–4098.
17. Bokes P, King JR, Loose M. A bistable genetic switch which does not require high co-operativity at the promoter: a two-timescale model for the PU.1-GATA-1 interaction. *Math Med Biol*. 2009;26(2):117–132.
18. Lai X, Schmitz U, Gupta SK, et al. Computational analysis of target hub gene repression regulated by multiple and cooperative miRNAs. *Nucleic Acids Res*. 2012;40(18):8818–8834.
19. Ratushny AV, Ramsey SA, Roda O, et al. Control of transcriptional variability by overlapping feed-forward regulatory motifs. *Biophys J*. 2008;95(8):3715–3723.
20. Vera J, Schmitz U, Lai X, et al. Kinetic modeling-based detection of genetic signatures that provide chemoresistance via the E2F1-p73/DNp-73-miR-205 network. *Cancer Res*. 2013;73(12):3511–3524.
21. Chatterjee MS, Denney WS, Jing H, et al. Systems biology of coagulation initiation: kinetics of thrombin generation in resting and activated human blood. *PLoS Comput Biol*. 2010;6(9):e1000950.
22. Goltsov A, Tashkandi G, Langdon SP, et al. Kinetic modelling of in vitro data of PI3K, mTOR1, PTEN enzymes and on-target inhibitors Rapamycin, BEZ235, and LY294002. *Eur J Pharm Sci*. 2017;97:170–181.
23. Groban ES, Clarke EJ, Salis HM, et al. Kinetic buffering of cross talk between bacterial two-component sensors. *J Mol Biol*. 2009;390(3):380–393.
24. Huang C, Wu M, Du J, et al. Systematic modeling for the insulin signaling network mediated by IRS(1) and IRS(2). *J Theor Biol*. 2014;355:40–52.
25. Lang CF, Maxian O, Anneken A, et al. Oligomerization and positive feedback on membrane recruitment encode dynamically stable PAR-3 asymmetries in the *C. elegans* zygote. *bioRxiv*. 2023.
26. Omelyanchuk LV, Munzarova AF. Theoretical model of mitotic spindle microtubule growth for FRAP curve interpretation. *BMC Syst Biol*. 2017;11(Suppl 1):378.
27. Reader PP, Shaw AM. Kinetic analysis of the multivalent ligand binding interaction between Protein A/G and IgG: A standard system setting. *J Phys Chem B*. 2017;121(38):8919–8925.
28. Diaz-Parga P, de Alba E. Inflammasome regulation by adaptor isoforms, ASC and ASCb, via differential self-assembly. *J Biol Chem*. 2022;298(3):101566.
29. Nägele T, Weckwerth W. Mathematical modeling reveals that metabolic feedback regulation of SnRK1 and hexokinase is sufficient to control sugar homeostasis from energy depletion to full recovery. *Front Plant Sci*. 2014;5:365.
30. Ryll A, Bucher J, Bonin A, et al. A model integration approach linking signalling and gene-regulatory logic with kinetic metabolic models. *BioSystems*. 2014;124:26–38.
31. Salim T, May EE. In silico kinetic model of iNOS expression in macrophages. *Conf Proc IEEE Eng Med Biol Soc*. 2014;2014:1159–1161.
32. Sier JH, Thumser AE, Plant NJ. Linking physiologically-based pharmacokinetic and genome-scale metabolic networks to understand estradiol biology. *BMC Syst Biol*. 2017;11(1):141.
33. Chung T, Chang I, Kim S. Development of equation of motion deciphering locomotion including omega turns of *Caenorhabditis elegans*. *eLife*. 2024;12:RP92562.
34. Larkin CI, Dunn MD, Shoemaker JE, et al. A detailed kinetic model of Eastern equine encephalitis virus replication in a susceptible host cell. *bioRxiv*. 2024.
35. Ravel G, Bergmann M, Trubuil A, et al. Inferring characteristics of bacterial swimming in biofilm matrix from time-lapse confocal laser scanning microscopy. *eLife*. 2022;11:e76513.
36. Cruz LAB, Hebly M, Duong G-H, et al. Similar temperature dependencies of glycolytic enzymes: an evolutionary adaptation to temperature dynamics? *BMC Syst Biol*. 2012;6:151.
37. Goßen J, Ribeiro RP, Bier D, et al. AI-based identification of therapeutic agents targeting GPCRs: introducing ligand type classifiers and systems biology. *Chem Sci*. 2023;14(32):8651–8661.
38. Millard P, Gosselin-Monplaisir T, Uttenweiler-Joseph S, et al. Acetate is a beneficial nutrient for *E. coli* at low glycolytic flux. *EMBO J*. 2023;42(15):e113079.
39. Su Y, Bintz M, Yang Y, et al. Phenotypic heterogeneity and evolution of melanoma cells associated with targeted therapy resistance. *PLoS Comput Biol*. 2019;15(6):e1007034.
40. Xu Y, Bergenstal RM, Dunn TC, et al. Addressing shortfalls of laboratory HbA1c using a model that incorporates red cell lifespan. *eLife*. 2021;10:e69456.
41. Gelbach PE, Zheng D, Fraser SE, et al. Kinetic and data-driven modeling of pancreatic β -cell central carbon metabolism and insulin secretion. *PLoS Comput Biol*. 2022;18(10):e1010555.
42. van Niekerk DD, du Toit F, Green K, et al. A detailed kinetic model of glycolysis in *Plasmodium falciparum*-infected red blood cells for antimalarial drug target identification. *J Biol Chem*. 2023;299(9):105111.
43. Pandey V, Sultan M, Kashofer K, et al. Comparative analysis and modeling of the severity of steatohepatitis in DDC-treated mouse strains. *PLoS One*. 2014;9(10):e111006.
44. Sen P, Vial HJ, Radulescu O. Kinetic modelling of phospholipid synthesis in *Plasmodium knowlesi* unravels crucial steps and relative importance of multiple pathways. *BMC Syst Biol*. 2013;7:123.
45. Van de Poel B, Bulens I, Hertog MLATM, et al. A transcriptomics-based kinetic model for ethylene biosynthesis in tomato (*Solanum lycopersicum*) fruit: development, validation and exploration of novel regulatory mechanisms. *New Phytol*. 2014;202(3):952–963.
46. Patel K, Barter M, Soul J, et al. Systems analysis of miR-199a/b-5p and multiple miR-199a/b-5p targets during chondrogenesis. *eLife*. 2024;12:RP89701.
47. Linke C, Chasapi A, González-Novo A, et al. A Clb/Cdk1-mediated regulation of Fkh2 synchronizes CLB expression in the budding yeast cell cycle. *NPJ Syst Biol Appl*. 2017;3:7.
48. Gilchrist M, Thorsson V, Li B, et al. Systems biology approaches identify ATF3 as a negative regulator of Toll-like receptor 4. *Nature*. 2006;441(7090):173–178.
49. Grundmann M, Tikhonova IG, Hudson BD, et al. A molecular mechanism for sequential activation of a G protein-coupled receptor. *Cell Chem Biol*. 2016;23(3):392–403.

50. Turcotte M, Tang W, Ross EM. Coordinate regulation of G protein signaling via dynamic interactions of receptor and GAP. *PLoS Comput Biol*. 2008;4(8):e1000148.
51. Croft W, Hill C, McCann E, et al. A physiologically required G protein-coupled receptor (GPCR)-regulator of G protein signaling (RGS) interaction that compartmentalizes RGS activity. *J Biol Chem*. 2013;288(38):27327–27342.
52. Kierzek AM, Zhou L, Wanner BL. Stochastic kinetic model of two component system signalling reveals all-or-none, graded and mixed mode stochastic switching responses. *Mol Biosyst*. 2010;6(3):531–542.
53. Edvinsson JM, Shah AJ, Palmer LG. Kir4.1 K⁺ channels are regulated by external cations. *Channels (Austin)*. 2011;5(3):269–279.
54. Arbona J-M, Goldar A, Hyrien O, Arneodo A, et al. The eukaryotic bell-shaped temporal rate of DNA replication origin firing emanates from a balance between origin activation and passivation. *eLife*. 2018;7:e35192.
55. Geyer T, Lauck F, Helms V. Molecular stochastic simulations of chromatophore vesicles from *Rhodobacter sphaeroides*. *J Biotechnol*. 2007;129(2):212–228.
56. Nielsen J, Petranovic D. Modeling for understanding and engineering metabolism. *QRB Discov*. 2025;6:e11.
57. Rangamani P, Iyengar R. Modelling cellular signalling systems. *Essays Biochem*. 2008;45:83–94.
58. Heijnen JJ, Verheijen PJT. Parameter identification of in vivo kinetic models: limitations and challenges. *Biotechnol J*. 2013;8(7):768–775.